TEMS Academy of Olympiad Math
Adhoc Algebra Problem Set

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Problem 1. Evaluate the sum

$$\sum_{k=1}^{n} \frac{k}{2^k}$$

Problem 2. Let x and y be real numbers such that $x - y = 4$ and $x^3 - y^3 = 28$. Find xy.

Problem 3. Find all real numbers x such that

$$(x^2 - 9x + 19)^{x^2-3x+2} = 1$$

Problem 4. Evaluate the sum

$$\sum_{k=1}^{n} k!(k^2 + k + 1)$$

Problem 5. Let a, b, c be real numbers such that

$$|a| \geq |b + c|, \qquad |b| \geq |c + a|, \qquad |c| \geq |a + b|$$

Prove that $a + b + c = 0$.

Problem 6. Evaluate the sum

$$\frac{3}{1! + 2! + 3!} + \frac{4}{2! + 3! + 4!} + \cdots + \frac{2027}{2025! + 2026! + 2027!}$$

Problem 7. Solve over $\mathbb{R}$:

$$x^2 - 4y + 7 = 0$$
$$y^2 - 6z + 14 = 0$$
$$z^2 - 2x - 7 = 0$$

Problem 8. Find all real numbers x for which

$$\frac{8^x + 27^x}{12^x + 18^x} = \frac{7}{6}$$

Problem 9. Evaluate the product

$$\prod_{k=2}^{2027} \frac{k^3-1}{k^3+1}$$

Problem 10. Let x and y be real numbers such that $x \neq 1$, $y \neq 1$ and

$$(x-1)^2 + (y-1)^2 = xy(x+y-3) + 1$$

Show that $x + y = 1$.

Problem 11. Find all real quadruples (a, b, c, d) satisfying

$$\begin{aligned}(b+c+d)^{2026} &= 3a\\ (a+c+d)^{2026} &= 3b\\ (a+b+d)^{2026} &= 3c\\ (a+b+c)^{2026} &= 3d\end{aligned}$$

Problem 12. Assign a positive integer to each face of a cube. At each vertex take the product of the labels on its three incident faces. These eight products sum to 1001. Determine the sum of the six face labels.

Problem 13. Let a, b, c, d and e be positive integers such that

$$abcde = a + b + c + d + e$$

Find the maximum possible value of $\max\{a, b, c, d, e\}$.

Problem 14. For $n \geq 1$ and $x \in \mathbb{R}$ with $\sin(2^n x) \neq 0$, prove that

$$\sum_{k=1}^{n} \frac{1}{\sin(2^k x)} = \cot x - \cot(2^n x)$$

Problem 15. Find all real quadruples (a, b, c, d) satisfying

$$\begin{aligned}abc + ab + bc + ca + a + b + c &= 1\\ bcd + bc + cd + db + b + c + d &= 9\\ cda + cd + da + ac + c + d + a &= 9\\ dab + da + ab + bd + d + a + b &= 9\end{aligned}$$

Problem 16. Define $a_k = \frac{1}{k^2+k}$ for $k \geq 1$. Positive integers $m < n$ satisfy

$$a_m + a_{m+1} + \cdots + a_{n-1} = \frac{1}{29}$$

Determine $m + n$.

Problem 17. Find all real numbers x such that

$$2^x + 3^x - 4^x + 6^x - 9^x = 1$$

Problem 18. Let $(x'_1, \ldots, x'_{2027})$ be a permutation of the real numbers $(x_1, \ldots, x_{2027})$. Suppose

$$x_i + x_{i+1} = 2x'_i$$

for $1 \leq i \leq 2027$ and $x_{2028} = x_1$. Prove that all x_i's are equal.

Problem 19. Evaluate

$$\frac{2}{4-1} + \frac{2^2}{4^2-1} + \frac{2^4}{4^4-1} + \frac{2^8}{4^8-1} + \cdots$$

Problem 20. Let x, y, z be positive reals such that

$$xyz = 1$$
$$x + \frac{1}{z} = 5$$
$$y + \frac{1}{x} = 29$$

Compute $z + \frac{1}{y}$.

Problem 21. Let $a_1 = a_2 = 1$ and $a_{n+2} = a_{n+1} + a_n$ for $n \geq 1$. Evaluate

$$\sum_{n=1}^{\infty} \frac{a_n}{4^{n+1}}$$

Problem 22. Find all real numbers x such that

$$\lfloor x \rfloor^3 - 7 \left\lfloor x + \frac{1}{3} \right\rfloor = -13$$

Problem 23. Find all real numbers x_1, x_2, x_3, x_4 satisfying

$$x_1 + x_2x_3x_4 = 2$$
$$x_2 + x_1x_3x_4 = 2$$
$$x_3 + x_1x_2x_4 = 2$$
$$x_4 + x_1x_2x_3 = 2$$

Problem 24. Let F_n be the nth Fibonacci number i.e. $F_0 = F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$. Evaluate the sum

$$\sum_{n=1}^{\infty} \frac{1}{F_{n-1}F_{n+1}}$$

Problem 25. A real sequence $(a_n)_{n\geq 1}$ satisfies

$$a_n = a_{n-1} + a_{n+2}$$

for $n \geq 2$. Determine the maximum number of consecutive terms which are all positive.

Problem 26. Let a, b and c be distinct nonzero real numbers such that

$$a + \frac{1}{b} = b + \frac{1}{c} = c + \frac{1}{a}$$

Prove that $|abc| = 1$.

Problem 27. Find all real numbers x such that

$$10^x + 11^x + 12^x = 13^x + 14^x$$

Problem 28. Determine all positive rational pairs (a, b) for which

$$\sqrt{a} + \sqrt{b} = \sqrt{2 + \sqrt{3}}$$

Problem 29. Find all real numbers x such that

$$\lfloor \lfloor x \rfloor x \rfloor = 2024$$

Problem 30. Determine all real triples (a, b, c) satisfying

$$\begin{aligned} a^3 + 3ab^2 + 3ac^2 - 6abc &= 1 \\ b^3 + 3ba^2 + 3bc^2 - 6abc &= 1 \\ c^3 + 3ca^2 + 3cb^2 - 6abc &= 1 \end{aligned}$$

Problem 31. Let $a = \frac{\pi}{2026}$. Determine the least positive integer n such that

$$2\sum_{k=1}^{n} \cos(k^2 a) \sin(ka)$$

is an integer.

Problem 32. Let x, y and z be complex numbers such that $x + y + z = 2, x^2 + y^2 + z^2 = 3$ and $xyz = 4$. Evaluate

$$\frac{1}{xy + z - 1} + \frac{1}{yz + x - 1} + \frac{1}{zx + y - 1}$$

Problem 33. Find all real numbers x such that

$$4x^2 - 40\lfloor x \rfloor + 51 = 0$$

Problem 34. Evaluate the sum

$$\sum_{k=1}^{\infty} \tan^{-1}\left(\frac{2}{k^2}\right)$$

Problem 35. Solve the system of equations

$$\begin{aligned} x^2y + 2 &= x + 2yz \\ y^2z + 2 &= y + 2zx \\ z^2x + 2 &= z + 2xy \end{aligned}$$

for real numbers x, y, z.

Problem 36. Evaluate the sum

$$\{\ln(1+e)\} + \{\ln(1+e^2)\} + \{\ln(1+e^4)\} + \{\ln(1+e^8)\} + \cdots$$

where $\{t\} = t - \lfloor t \rfloor$.

Problem 37. Solve the system of equations

$$\begin{aligned} (x_1 - x_2 + x_3)^2 &= x_2(x_4 + x_5 - x_2) \\ (x_2 - x_3 + x_4)^2 &= x_3(x_5 + x_1 - x_3) \\ (x_3 - x_4 + x_5)^2 &= x_4(x_1 + x_2 - x_4) \\ (x_4 - x_5 + x_1)^2 &= x_5(x_2 + x_3 - x_5) \\ (x_5 - x_1 + x_2)^2 &= x_1(x_3 + x_4 - x_1) \end{aligned}$$

for real numbers x_1, x_2, x_3, x_4, x_5.

Problem 38. Evaluate the sum

$$\sum_{n=2}^{\infty} \frac{n^4 + 3n^2 + 10n + 10}{2^n(n^4 + 4)}$$

Problem 39. Find all real triples (x, y, z) satisfying

$$\begin{aligned} 6x - y + z^2 &= 3 \\ x^2 - y^2 - 2z &= -1 \\ 6x^2 - 3y^2 - y - 2z^2 &= 0 \end{aligned}$$

Problem 40. Evaluate the sum

$$\binom{2027}{2} + \binom{2027}{5} + \binom{2027}{8} + \cdots + \binom{2027}{2027}$$

Problem 41. Find all real numbers x and with $x > y$ satisfying

$$x^2y^2 + x^2 + y^2 + 2xy = 40$$
$$xy + x + y = 8$$

Problem 42. Find all positive real numbers x satisfying

$$5x \max\left\{8x, \frac{1}{8x}\right\} \max\left\{25x, \frac{1}{25x}\right\} = 1$$

Problem 43. Find all real quadruples (a, b, c, d) satisfying

$$ab + c + d = 3$$
$$bc + d + a = 5$$
$$cd + a + b = 2$$
$$da + b + c = 6$$

Problem 44. Do there exist pairwise distinct rational numbers x, y, z such that

$$\frac{1}{(x-y)^2} + \frac{1}{(y-z)^2} + \frac{1}{(z-x)^2} = 2026?$$

Problem 45. Let a, b, c, d be positive real numbers satisfying

$$a^2 + d^2 - ad = b^2 + c^2 + bc$$
$$a^2 + b^2 = c^2 + d^2$$

Find all possible values of

$$\frac{ab + cd}{ad + bc}$$

Problem 46. Find all positive real triples (a, b, c) satisfying

$$\lfloor a \rfloor bc = 3$$
$$a\lfloor b \rfloor c = 4$$
$$ab\lfloor c \rfloor = 5$$

Problem 47. Does there exist an angle $\alpha \in (0, \pi/2)$ such that

$$\sin\alpha, \quad \cos\alpha, \quad \tan\alpha, \quad \cot\alpha$$

can be arranged as four consecutive terms of an arithmetic progression?

Problem 48. Let x and y be real numbers such that

$$x = y(3-y)^2, \qquad y = x(3-x)^2$$

Find all possible values of $x + y$.

Problem 49. Determine the number of integers n satisfying

$$1 + \left\lfloor \frac{100n}{101} \right\rfloor = \left\lceil \frac{99n}{100} \right\rceil$$

Problem 50. Find all real triples (x, y, z) satisfying

$$x^2y + y^2z + z^2 = 0$$
$$z^3 + z^2y + zy^3 + x^2y = \frac{1}{4}(x^4 + y^4)$$